

MADHU LATHA PULIMI

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Open to Remote ◇ Open to Relocate ◇ Open to Hybrid ◇ Open to Travel ◇ Open to On-Site

PROFESSIONAL SUMMARY

- Extensive experience in Data Engineering across finance (**Bank of America**), automotive (**Renault/RNTBCI**), retail (**Target/Wipro**), and healthcare (**Cognizant, AmeriHealth – health insurance provider**), designing and optimizing enterprise ETL pipelines and data architectures, improving reliability, reducing latency, enhancing data quality, and solving complex large-scale data engineering problems.
- Hands-on experience with Big Data technologies including **Apache Spark, Hive, PySpark, Scala, and Hadoop**, working on high-volume data ingestion pipelines processing more than **100M records per cycle**, optimizing partition strategies, execution plans, and parallelism at scale.
- **Optimized data processing configurations** to improve throughput, reduce job execution times, and eliminate performance bottlenecks across enterprise Data Lake environments.
- **Hands-on with Azure Databricks and Snowflake** for cloud-based **data warehousing** and pipeline orchestration, leveraging **Databricks notebooks, Delta Lake**, and Snowflake's elastic compute to dramatically improve query performance, also delivered hybrid cloud-to-on-premises **ETL architectures** using **Azure Data Factory** and **Azure Blob Storage** to reduce operational overhead and enable seamless data movement.
- **Data & AI Expert** with extensive experience building distributed pipelines processing more than **100M records per cycle**.
- Specialist in high-performance **RAG systems** and **Agentic AI** orchestration, integrating **Vector Databases (FAISS)** and **Knowledge Graphs (Neo4j)** to ensure contextual accuracy.
- Engineered automated **Data Factory pipelines** in Microsoft Fabric to ingest and transform large-scale datasets from multiple sources, enabling clean, model-ready data for machine learning workflows.
- Strong background in **AI/ML and NLP**, including **GPT-4, BERT, Transformers, LSTM, CNN, LangChain, RAG**, and **FAISS**, with proven project experience in building fake news classifiers with more than **92% accuracy** using **PyTorch**, designing **GPT-4-powered RAG-based QA systems** with **semantic search**, and deploying **Streamlit applications** integrated with **Azure Functions** for **scalable real-time inference**.
- Built ML pipeline automation and batch inference systems using **PySpark, Apache Airflow**, and **dbt**, processing more than **100M records per cycle**, benchmarking **XGBoost, Decision Tree, Random Forest**, and **Logistic Regression** models optimized via **GridSearchCV**, reducing **pipeline failure rates** and enabling faster model refresh cycles for targeted rewards campaigns and **LLM-based rule generation workflows**.
- **Currently building an Agentic AI system** using a **multi-agent MCP architecture** for automated **subscription detection and financial intelligence** from email and SMS data, combining **NLP-based entity extraction, LLM-powered reasoning**, and **real-time billing prediction** to deliver proactive financial alerts without manual intervention.
- Proven **team player** and **independent contributor** across the full **software development lifecycle (SDLC)**, from **requirements engineering** through **CI/CD deployment**. Experienced in collaborating with distributed teams, conducting code reviews, and partnering with data scientists and business stakeholders across **financial, automotive, and retail domains** to **translate complex model outputs into actionable business strategies**.

PUBLICATIONS & AWARDS

- **SEAMS 2026:** "Grammar-Constrained Refinement of Safety Operational Rules Using Language in the Loop: What Could Go Wrong?" **Published in the Proceedings of the 21st International Symposium on Software Engineering for Adaptive and Self-Managing Systems.**
- **Hackathon:** Accessible Mail: GenAI-Powered Email Accessibility — 2nd Place, Michigan ITS HackwithFriends
- **Poster Presentation:** Won **2nd Place** at the **NLP @Michigan** competition among 60 posters, presenting research on [Can Rule-Constrained Reasoning Ensure TrustWorthiness in LLMs?]

TECHNICAL SKILLS

- **Data Engineering & Big Data:** Apache Spark (PySpark/Scala), Hadoop Ecosystem (HDFS, Hive, MapReduce), Python, Unix Shell Scripting, Apache Airflow, dbt (data build tool), Delta Lake.
- **Cloud & Platforms:** Azure Databricks, Snowflake, Azure Machine Learning (Azure ML), Azure Data Factory (ADF), Azure Blob Storage, ADLS Gen2.
- **Generative AI & Advanced ML:** Transformer Architectures, Large Language Models (GPT-4, Claude, BERT), Agentic AI Systems, Multi-Agent Orchestration, Retrieval-Augmented Generation (RAG), Prompt Engineering, Fine-Tuning, Quantization (4-bit/8-bit), Embeddings, Vector Databases (FAISS, Pinecone, Milvus), LangChain, LangGraph.
- **Core AI / ML Foundations:** Natural Language Processing (NLP), Sequence Modeling (LSTM, GRU), Reinforcement Learning (Project), Autonomous Systems, PyTorch, TensorFlow, Scikit-learn.
- **Databases & ETL Tools:** Oracle, Teradata, DB2, Vertica, SQL Server, MySQL, IBM InfoSphere DataStage, SSIS, Data Warehousing, Dimensional Modeling (Star/Snowflake Schema).
- **DevOps & Version Control:** Git (GitHub/Bitbucket), Docker, Kubernetes, Jenkins, CI/CD Pipelines, MLOps (MLflow, Weights & Biases).
- **BI & Visualization:** Tableau, QlikView, Streamlit.
- **Workflow & Governance:** Control-M, Autosys, JIRA, Erwin Data Modeler.

MASTER'S THESIS

Integrating LLMs into Cyber-Physical Systems (CPS) requires transforming probabilistic AI outputs into verifiable safety specifications to prevent unsafe physical behaviors. This research investigates how a grammar- and rule-constrained framework, utilizing structural enforcement, semantic validation, and counterfactual grounding, can enhance the trustworthiness and interpretability of LLM-driven operational rule refinement. Building on findings from SEAMS 2026, the study hypothesizes that these principled constraints allow LLMs to produce domain-aware, traceable, and formally verifiable requirements. Ultimately, this approach repositions LLMs from opaque generators of potentially unsafe specifications to accountable assistants capable of meeting the rigorous safety and resilience standards essential for real-world CPS deployments.

PROFESSIONAL EXPERIENCE

Bank of America(Full-Time)-Onsite(BA Continuum India Pvt Ltd)

Jan 2017 – Nov 2023

<https://www.bankofamerica.com/>

- Architected scalable data pipelines using PySpark, Scala, Spark SQL and Hive process 100M+ records for model training, batch inference, and large-scale transformation, loaded curated datasets into Delta la following data warehousing best practices.
- Executed the migration of AML workflows from Unix-based environment to a scalable Hadoop ecosystem, including data migration from Vertica to Teradata, re-architected shell scripts into PySpark applications, and developed automation scripts to streamline repetitive tasks, significantly reducing processing time and enabling handling of multi-terabyte datasets previously beyond legacy capacity.
- Developed modular functional components within the CRH to detect sophisticated Money Laundering "red flags", translated intricate regulatory requirements into high-performance transformation modules to ensure enterprise-level compliance readiness.
- Designed generic inbound/outbound data processing frameworks within enterprise Data Lake environments, engineered for distributed processing of high-volume transactional and behavioral data using PySpark.
- Built a cross-system data validation tool to verify integrity between Hive and Teradata, developed automated alerts, audit reports, and failure notification logic that improved production reliability and reduced incident recurrence.
- • Developed an end-to-end ML POC for Rewards Optimization, benchmarking **XGBoost and Random Forest** classifiers to drive targeted engagement.
- Drove \$2M–\$5M in cost savings by redesigning pipeline execution sequences, eliminating redundant data movements, and optimizing job scheduling across the Data Lake, significantly reducing infrastructure and compute overhead.
- Managed requirements-to-deployment cycles within a CI/CD environment, acted as a lead escalation point for data quality failures and batch delays under tight production SLAs, delivering root-cause fixes that stabilized the system.
- Designed and optimized complex data schemas for identity resolution, leveraging graph-based logic to map entity relationships and detect hidden connections in high-volume financial transaction data for AML monitoring.
- Ensured strict adherence to PII Governance and Data Residency protocols while handling sensitive customer information, implementing robust data masking and access controls to comply with global financial regulatory frameworks, including GLBA and AML standards.
- Developed an end-to-end ML POC for Rewards Optimization using CRH data, engineered behavioral feature pipelines and benchmarked classifiers (XGBoost, Random Forest) to drive targeted customer engagement campaigns.

RNTBCI (Renault Subsidiary)(Full-Time)-Onsite

Jan 2015 – Oct 2016

<https://rntbci.in/>

- Designed and implemented complex ETL workflows using IBM DataStage and Apache Airflow to populate a centralized Data Warehouse, processing more than 50M records with consistent data integrity, engineered SCD Type 2 dimensions and comprehensive source-to-target mappings to enable historical tracking and time-series analysis across vehicle supply chain metrics.
- Built a UNIX-based Control Module for dynamic monitoring, validation, and automated rerunning of ETL jobs, leveraged Azure Data Factory for orchestration between on-premises systems and cloud storage and

utilized Azure Blob Storage as a staging layer for large-volume raw data, reducing manual intervention and improving pipeline scheduling efficiency.

- Contributed to cloud migration by integrating Azure services alongside existing IBM DataStage infrastructure, laying the groundwork for a hybrid ETL architecture that improved scalability and reduced operational overhead.
- Performed data profiling and exploratory analysis on Renault's vehicle production and supply chain datasets, identifying anomalies, trend patterns, and quality gaps that directly informed BI dashboards and management-level decision-making.
- Led offshore team coordination, conducted code reviews, and managed release planning to ensure timely, high-quality delivery.

Cognizant- amerihealth caritas (Full-Time)-Onsite.

Aug 2013 – Jan 2015

<https://www.cognizant.com/in/en>

- Developed and optimized **IBM DataStage ETL pipelines** for high-volume healthcare claims and patient data processing, implementing modular **shared containers**, **PL/SQL scripts**, and data validation frameworks to ensure performance, reusability, and data quality.
- Built data standardization and conversion modules to align heterogeneous healthcare data formats with **enterprise data models (EDM)** and compliance requirements.

Wipro Technologies – Target Corporation (Full-Time)-Onsite

Aug 2009 – Aug 2013

<https://www.wipro.com/>

- Designed and implemented scalable **ETL pipelines** for POS, supply chain, and inventory data into **Teradata, DB2, and Oracle data warehouses**, leveraging **Teradata FastLoad/MultiLoad** for high-volume data ingestion and performance optimization.
- Automated end-to-end data operations using **UNIX shell scripting** and **Autosys**, and designed enterprise **ER models** using **Erwin Data Modeler** to support robust data warehouse architecture and operational efficiency.

ACADEMIC & AI PROJECTS

University of Michigan – Dearborn | Research Assistant

<https://umdearborn.edu/>

AI & Machine Learning | Deep Learning & Applied AI Project.

- Developed **CNN and LSTM models in PyTorch** for fake news classification using **TF-IDF and Word2Vec embeddings**, achieving greater than **92% accuracy**, with a real-time **Streamlit inference app** for deployment.
- Built **RAG-based QA systems** using GPT-4, LangChain, and FAISS to enable semantic search across large document collections. Additionally, implemented **MLP (Multi-Layer Perceptron)** and **Ridge Regression** models for digit recognition and pricing optimization, leveraging **cross-validation** and **grid search** for hyperparameter tuning.
- Designed a **voice-activated passive Smart Assistant** that detects user departure via Wi-Fi state changes and triggers **context-aware reminders and notifications**, with NLP-based intent mapping for personalized checklist generation.

- Built an end-to-end pipeline for **autonomous vehicle safety rule generation**, using **grammar-constrained LLMs** to create validated, structured safety rules from sensor data, refined via **human-in-the-loop evaluation** (aligned with SEAMS 2026 research).
- Architected a multi-agent **MCP-based Agentic AI system** for subscription detection and financial intelligence, developed a **Neo4j-based** to map complex relationships between extracted financial entities (Email/SMS data) real time detection of recurring subscriptions, detect anomalies, and generate billing predictions using **LLM-driven reasoning**, **NLP-based information extraction**, and financial analytics.

EDUCATION

- **University of Michigan – Dearborn**-Master of Science in Artificial Intelligence- (Jan 2025- Apr 2026)
- **Sri Venkateswara University, Tirupati**-Bachelor of Technology in Mechanical Engineering- (Aug 2004- Apr 2008)